Artificial Neural Networks

The foundation of deep learning and modern AI — inspired by the biological brain.

Where ANNs Show Up

Neural networks power some of the most transformative technologies of our time.

Speech Recognition

Converting spoken language into text in real time.

Image Classification

Identifying objects, faces, and scenes in photos.

Autonomous Vehicles

Perceiving and navigating the physical world.

Medical Diagnosis

Detecting disease from scans and patient data.

Network Architecture

A neural network is a collection of interconnected neurons organised in layers – like an assembly line.

1

Input Layer

Receives raw data – pixel values, features – and distributes it forward without processing.

2

Hidden Layers

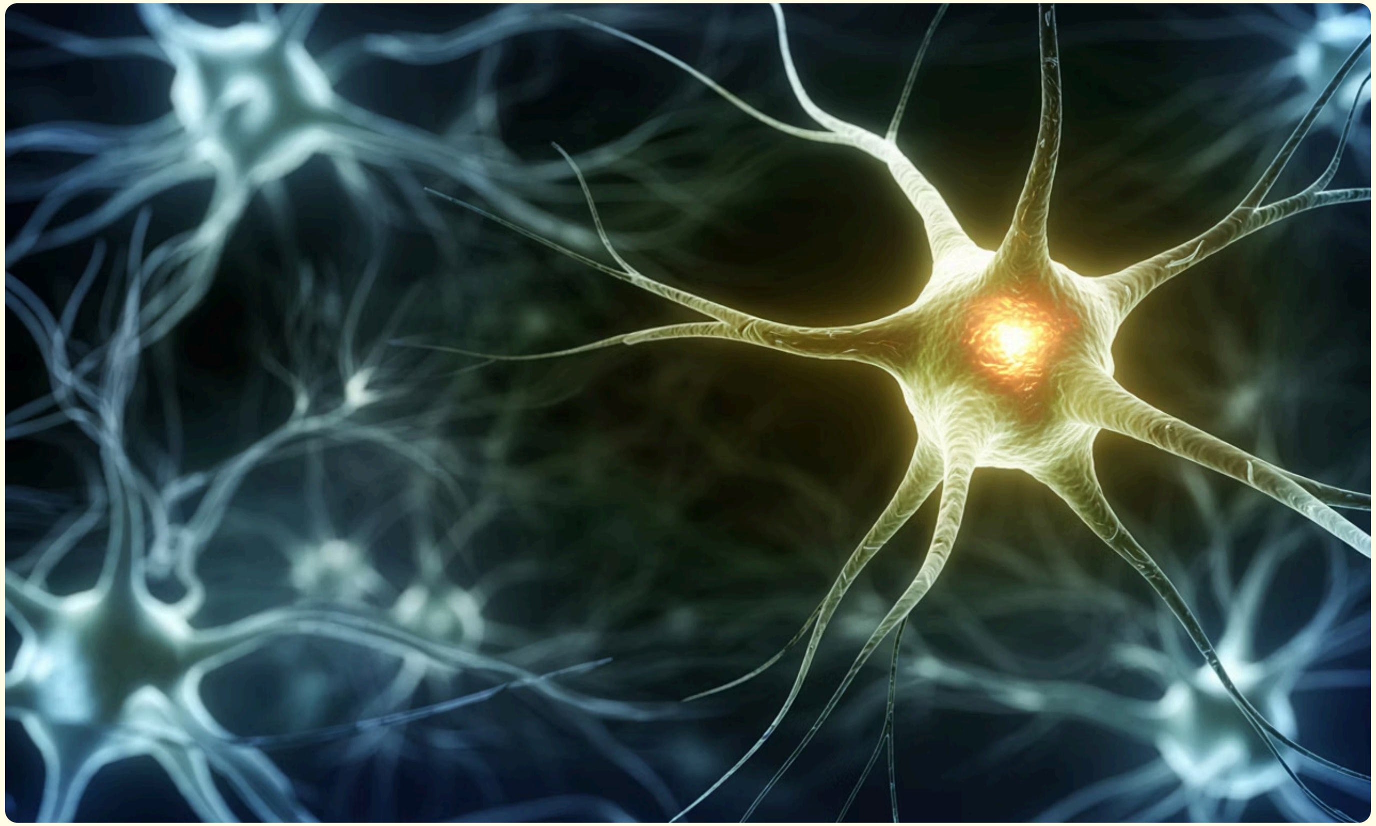
Where computation happens. Each layer extracts increasingly complex features. More layers = "deep" learning.

3

Output Layer

Produces the final result – a classification, prediction, or generated text.

The Artificial Neuron



How One Neuron Works

01

Receives multiple **input signals** from the previous layer

02

Computes a **weighted sum** of inputs plus a bias term

03

Passes the result through an **activation function**

04

Sends an **output signal** to the next layer

The Key Equation

Every neuron performs this simple calculation — yet millions working together create remarkable intelligence.

$$z = \sum(w_i \cdot x_i) + b$$

$$a = f(z) \quad \text{where } f \text{ is the activation function}$$

Inputs *x*

Raw data or signals from the previous layer — analogous to dendrites.

Weights *w*

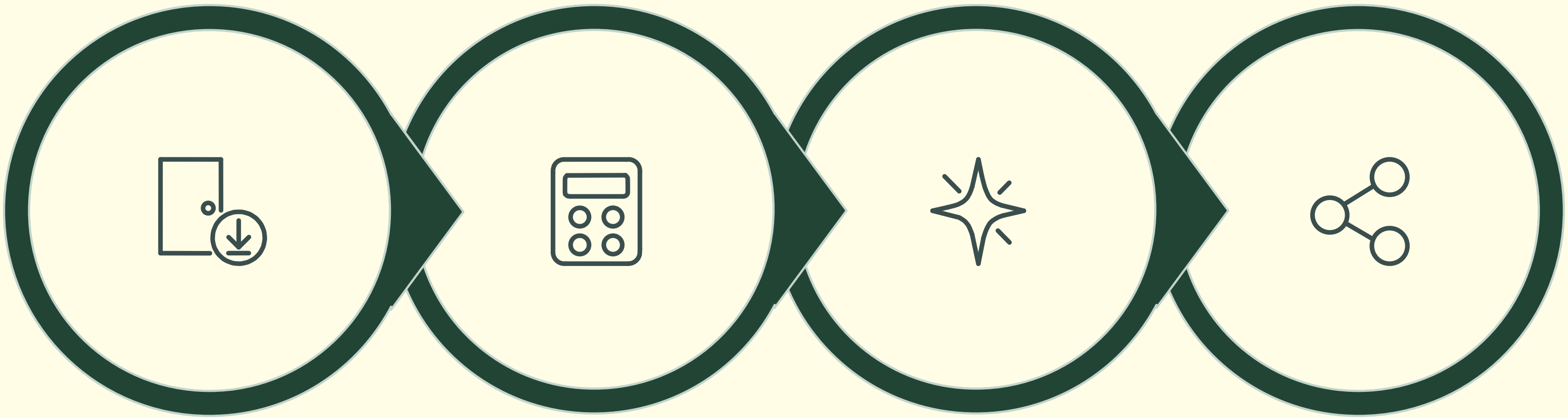
Strength of each connection. This is where the network's knowledge lives.

Bias *b*

Shifts the activation threshold, giving the neuron flexibility.

The Forward Pass

The **inference phase** — how the network turns raw input into a prediction, layer by layer.



Input Values

Weighted Sum

Activation

Output

Without activation functions, stacking layers would collapse into a single linear model. **Non-linearity** is what allows networks to learn complex, real-world relationships.

Activation Functions

Activation functions decide how much signal passes forward and introduce the non-linearity that makes deep learning powerful.

Function	Output Range	Strengths	Weakness	Use Case
Sigmoid	(0, 1)	Smooth, probabilistic	Vanishing gradients	Binary classification
Tanh	(-1, 1)	Zero-centred	Vanishing gradients	Recurrent networks
ReLU	[0, ∞)	Fast, scalable	Dying ReLU	Most hidden layers

Modern networks favour **ReLU and its variants** — Leaky ReLU, GELU, and Swish — for speed and stability.

What Activation Functions Do

Model Complex Patterns

Enable the network to learn non-linear relationships in data.

Control Information Flow

Determine which signals are amplified or suppressed.

Universal Approximation

Make neural networks capable of approximating any function.

Synapses & Learning

Weights determine the **strength and direction** of connections – excitatory or inhibitory. The network's knowledge is stored entirely in these values.

1

Forward Pass

Make a prediction from current weights.

2

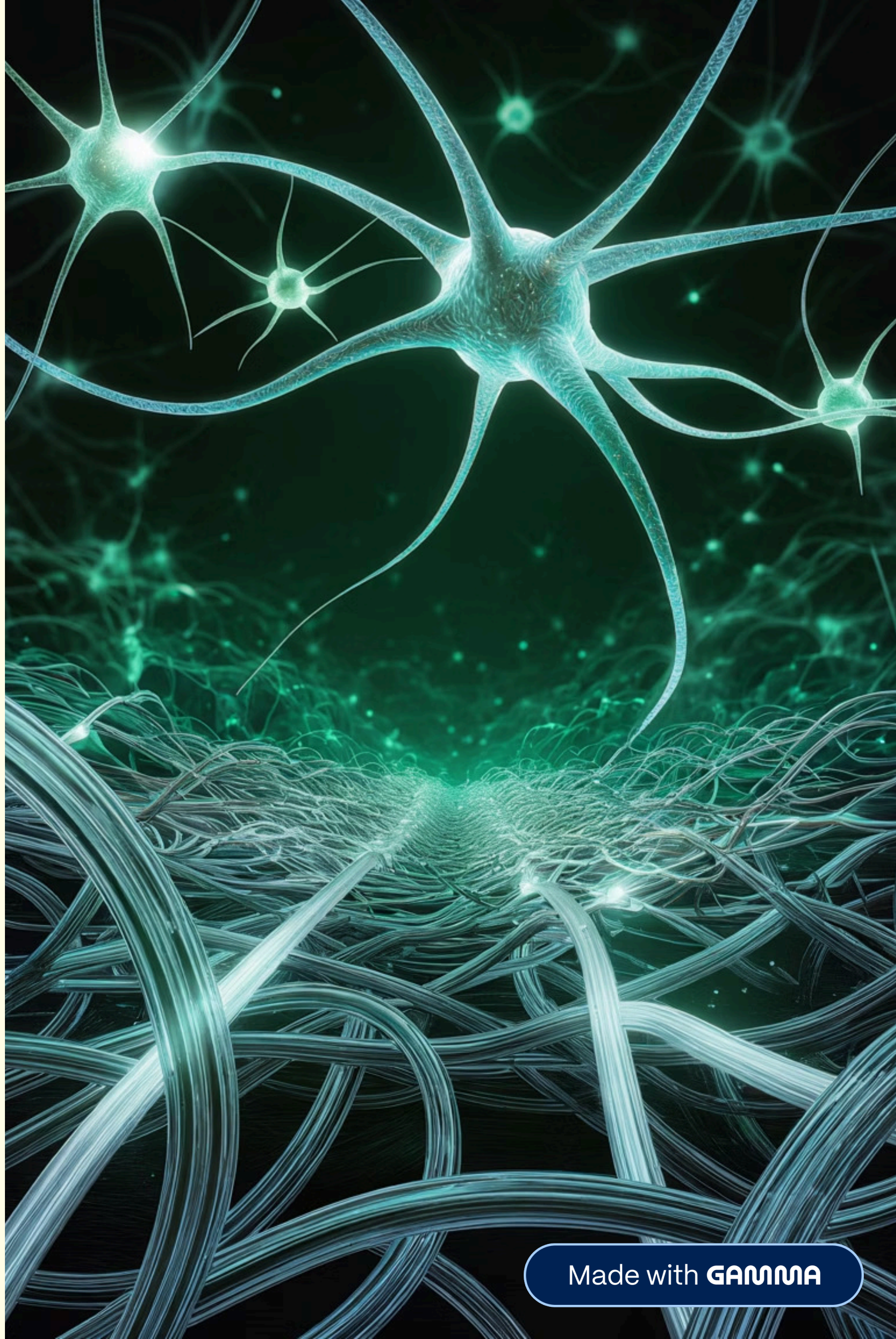
Calculate Loss

Measure the error between prediction and truth.

3

Backpropagate

Update weights via gradient descent to reduce error.



Key Takeaways

Inspired by Biology

ANNs mimic neurons but are ultimately **mathematical function approximators**.

Core Unit

The artificial neuron: **weighted sum + bias + activation function**.

Architecture

Input → Hidden Layers → Output. More hidden layers = deeper learning.

Forward Pass

How predictions are made — layer by layer, non-linearity at every step.

Weights = Knowledge

Learning is stored in synaptic weights, refined through backpropagation.

Next session: Training in depth — backpropagation, optimisers, and loss functions.